HOW MUCH DSA

DO YOU NEED FOR JAVA JOBS?





Tech | Career | Resources
@akanksha_buchke

SWIPE <<<

Save this

YOUR DSA PRIORITY MAP FOR JAVA JOBS

Must Know: 70% OF ALL INTERVIEW QUESTIONS

- Arrays & Strings
- HashMap / HashSet
- Stack & Queue
- Two Pointers
- Sliding Window
- Binary Search

GOOD TO KNOW: ~25% - ASKED AT MID/SENIOR LEVEL

- Binary Trees
- BFS / DFS
- Recursion + Backtracking
- Linked List
- Basic DP

SKIP FOR MOST JOBS: ~5% - ONLY FAANG+/ COMPETITIVE ROUNDS

- Segment Trees
- TriesBit Manipulation
- Advaneed DP
- Network Flow



ARRAYS & STRINGS 40% OF ALL ROUNDS.

The most asked topic in Java interviews - by a large margin. Not because it's easy, but because it tests real thinking.

Two Pointer Pattern:
Pair sum, remove duplicates, container with most water - all two pointers. Learn the template once, apply everywhere.
O(n) time · O(1) space

Sliding Window:
Longest substring, max sum subarray, minimum window - all one pattern. Interviewers love this because 80% of candidates still brute-force it.
replaces O(n2) with O(n)

Follow and comment "DSA" for PDF🙌🏻

→

HASHMAP & HASHSET THE SILENT PROBLEM-SOLVER.

Almost every array/string problem can be reduced to O(n) using a HashMap. If you're not reaching for it first, you're leaving points on the table.

Frequency Counter Pattern:
Anagram check, duplicate detection, group by count - all HashMap. Build frequency map in one pass, query in O(1).
Asked in: Amazon, Infosys, TCS Digital

Know Java's HashMap Internals:
Interviewers ask: "What happens on hash collision?" Know about chaining, load factor (0.75), and treeification at 8 entries. This alone separates you.
loadFactor . threshold . treeify @ 8



BINARY TREES & GRAPHS
KNOW THESE 3 TRAVERSALS.

You don't need to master every tree problem. You need to own the 3 traversals + BFS/DFS deeply. Everything else is built on top.

Master Inorder, Preorder, Postorder:
Not just recursively - know the iterative version with a Stack too. Interviewers often ask "can you do this without recursion?" - most can't.
Inorder of BST = sorted array

BFS = Level Order = Queue:
Shortest path, level by level problems, graph connected components - all BFS with a Queue. One pattern, 10+ problems solved.
Asked in: Flipkart, Uber, Zomato

STACK & QUEUE MOST UNDERRATED, MOST ASKED

Candidates skip Stack/Queue thinking it's too basic. Then they fail "valid parentheses" in round 1. It's not basic - it's a pattern engine.

Monotonic Stack - Learn This:
Next Greater Element, Daily Temperatures, Largest Rectangle in Histogram - all one pattern. Most devs have never heard of it. You'll stand out immediately.
O(n) solution to O(n2) problems

Java: Deque over Stack:
Java's Stack class is legacy and slow. Use ArrayDeque as a Stack in interviews. This detail shows you know Java - not just algorithms.
ArrayDeque · push/pop · peekFirst

STOP WASTING TIME ON THESE TOPICS.

Unless you're targeting Google, Meta, or top quant firms - these topics won't show up. Most Java job seekers burn weeks here for zero ROI.

Segment Trees & Fenwick Trees:
Range query problems. Appear in competitive programming. Almost never in Java backend job interviews. Skip unless you're targeting FAANG.
Only ~3% of Java job rounds

Advanced DP (Knapsack variants, Bitmask DP):
Basic DP (Fibonacci, Climbing Stairs, Coin Change) is enough for 95% of Java jobs. Bitmask DP is basically never asked outside FAANG.
Basic DP only - memo + tabulation

Follow and comment “DSA” for PDF🙌



AKANKSHA BUCHKE

SENIOR SOFTWARE ENGINEER @PAYPAL

EX @INTUIT | EX @TCS